

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

SS TAKEN IN ONENOTE

Department of Computer Science and Engineering (CSE)**QUIZ-2****WINTER SEMESTER, 2022-2023****DURATION: 35 MINUTES****FULL MARKS: 15****Math 4543: Numerical Methods****Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).**Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.**STUDENT ID:**

1. The Taylor series for a function
- $f(x)$
- is –

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

6
(CO1)
(PO2)

Using the Taylor series (with at least the first 4 terms), derive the Maclaurin series of –

(i) $\ln(1+x)$

(ii) $\ln(1-x)$

Then use these two series to determine the Maclaurin series of $\ln\left(\frac{1+x}{1-x}\right)$.**Solution:**The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.We know, $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$, and $\frac{d}{dx}(x^n) = nx^{n-1}$.

- (i) Now, for
- $f(x) = \ln(1+x)$
- , let's calculate the derivatives at the point
- $x = 0$
- .

$$f(x) = \ln(1+x), f(0) = \ln(1+0) = \ln(1) = 0; f'(x) = \frac{1}{1+x}, f'(0) = \frac{1}{1+0} = 1;$$

$$f''(x) = -\frac{1}{(1+x)^2}, f''(0) = -\frac{1}{(1+0)^2} = -1; f'''(x) = \frac{2}{(1+x)^3}, f'''(0) = \frac{2}{(1+0)^3} = 2;$$

Using the first 4 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \dots$$

For $x = 0$,

$$f(0+h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + \dots$$

$$\Rightarrow f(h) = 0 + h - \frac{h^2}{2!} + \frac{2h^3}{3!} + \dots$$

$$\Rightarrow f(x) = 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots$$

$$\Rightarrow \ln(1+x) = 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots$$

$$\Rightarrow \ln(1+x) = x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots$$

(ii) **Short version:** Here,

$$\begin{aligned}f(x) &= \ln(1 - x) \\&= \ln(1 + (-x)) \\&= (-x) - \frac{(-x)^2}{2!} + \frac{2(-x)^3}{3!} + \dots \quad [\text{using the Maclaurin series of } \ln(1 + x) \text{ from (i)}] \\&= -x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots\end{aligned}$$

Long version: Now, for $f(x) = \ln(1 - x)$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = \ln(1 - x), f(0) = \ln(1 - 0) = \ln(1) = 0; f'(x) = -\frac{1}{1 - x}, f'(0) = -\frac{1}{1 - 0} = -1;$$

$$f''(x) = -\frac{1}{(1 - x)^2}, f''(0) = -\frac{1}{(1 - 0)^2} = -1;$$

$$f'''(x) = -\frac{2}{(1 - x)^3}, f'''(0) = -\frac{2}{(1 - 0)^3} = -2;$$

Using the first 4 terms of the Taylor series,

$$f(x + h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \dots$$

For $x = 0$,

$$f(0 + h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + \dots$$

$$\Rightarrow f(h) = 0 - h - \frac{h^2}{2!} - \frac{2h^3}{3!} + \dots$$

$$\Rightarrow f(x) = 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots$$

$$\Rightarrow \ln(1 - x) = 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots$$

$$\Rightarrow \ln(1 - x) = -x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots$$

We know, $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$.

So,

$$\begin{aligned}\ln\left(\frac{1 + x}{1 - x}\right) &= \ln(1 + x) - \ln(1 - x) \\&= \left(x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots\right) - \left(-x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots\right) \\&= 2x + \frac{4x^3}{3!} + \dots\end{aligned}$$

Rubric:

- 2.5 points for the correct series of $\ln(1 + x)$.
- 2.5 points for the correct series of $\ln(1 - x)$.
- 1 point for the correct series of $\ln\left(\frac{1 + x}{1 - x}\right)$.

2. Compare and contrast the following interpolation methods— **Direct** method, **Newton’s Divided Difference** method, and **Lagrangian** method. 2
(CO3)
(PO2, PO4)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Direct	Newton’s Divided Difference	Lagrangian
Interpolant Polynomial Representation	$f_n(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$	$f_n(x) = b_0 + b_1(x - x_0) + \cdots + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$	$f_n(x) = \sum_{i=0}^n \left(\prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j} \right) f(x_i)$
Computational Complexity	Depends on the algorithm used to solve the equations. It may be $O(n^3)$, $O(n^3 \log(\ A\ + \ b\))$, or $O((n + 1)!)$.	Using memoized dynamic programming, we get a complexity of $O(n^2)$.	The naïve implementation is $O(n^2)$, but it can be optimized further down to $O(n \log^2(n))$.
Property of the terms	Each term has a coefficient a_i and these coefficients are obtained by solving all the equations simultaneously.	Each term has a coefficient b_i and each of these coefficients can be obtained by solving one equation.	The polynomial is a weighted sum of the given functional values $f(x_i)$, and each term is a product of the Lagrangian weight function $L_i(x)$ and the functional value $f(x_i)$.
Vulnerable to the Runge Phenomenon?	Yes	Yes	Yes

Rubric:

- 1 point for each valid comparison.

3. The specific heat capacity C of a substance is defined as the amount of heat that is required to raise the temperature of unit mass of that particular substance by 1 degree. Suppose, you are conducting an experiment to determine how much heat is required to bring some volume of water to its boiling point. The values of specific heat capacity C of water that you have calculated at different temperature values T are shown in the table below –

Temperature, T ($^{\circ}C$)	Specific Heat, C ($Jkg^{-1}^{\circ}C^{-1}$)
22	4181
42	4179
52	4186
82	4199
100	4217

Table 1: Specific heat C of water as a function of temperature T .

Determine the value of the specific heat at $T = 61^{\circ}C$ using the **Direct method** of interpolation and a second order polynomial.

Solution:

For second order polynomial interpolation (also called quadratic interpolation), the equation for the specific heat would be –

$$C(T) = a_0 + a_1T + a_2T^2$$

Here, the order $n = 2$. Now, let's choose the nearest $n + 1 = 2 + 1 = 3$ points that bracket the point $T = 61^{\circ}C$ and evaluate the polynomial. The 3 points are $T_0 = 42$, $T_1 = 52$, and $T_2 = 82$. The respective specific heat values are $C(T_0) = 4179$, $C(T_1) = 4186$, and $C(T_2) = 4199$. Using these values, we obtain the following equations –

$$C(42) = a_0 + a_1 \times 42 + a_2 \times (42)^2 = 4179$$

$$C(52) = a_0 + a_1 \times 52 + a_2 \times (52)^2 = 4186$$

$$C(82) = a_0 + a_1 \times 82 + a_2 \times (82)^2 = 4199$$

Writing these equations in matrix form, we have –

$$\begin{aligned} \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} \frac{533}{50} & -\frac{287}{25} & \frac{91}{50} \\ -\frac{67}{200} & \frac{31}{75} & -\frac{47}{600} \\ \frac{1}{400} & -\frac{1}{300} & \frac{1}{1200} \end{bmatrix} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4135 \\ 1.3267 \\ -6.6667 \times 10^{-3} \end{bmatrix} \end{aligned}$$

Hence,

$$C(T) = 4135 + 1.3267T - 6.6667 \times 10^{-3}T^2; \quad 42 \leq T \leq 82$$

At the point $T = 61$, we get,

$$C(61) = 4135 + 1.3267 \times 61 - 6.6667 \times 10^{-3} \times (61)^2 = 4191.2 Jkg^{-1}^{\circ}C^{-1}$$

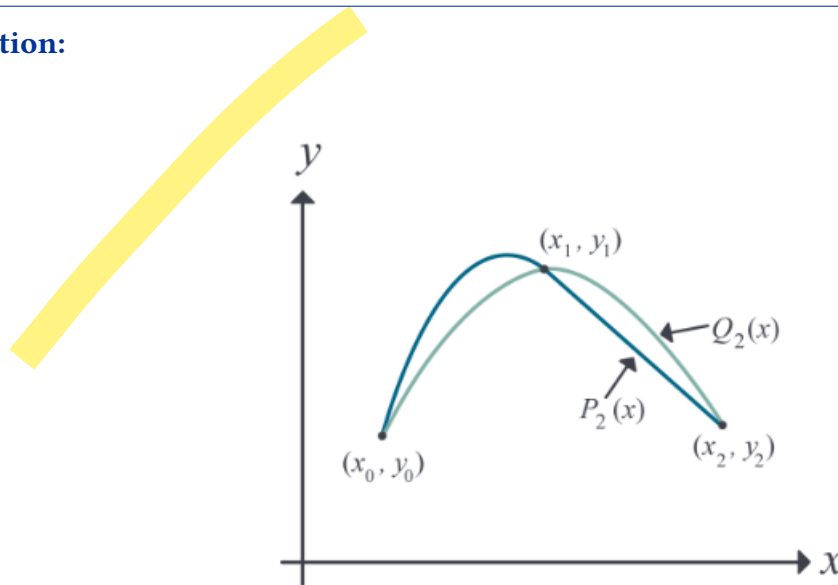
Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 1 point for the correct coefficient values.
- 1 point for the correct answer.

4. Prove that a polynomial of degree n or less that passes through $n + 1$ data points is unique.

3
(CO1)
(PO2)

Solution:



Proof: Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order n or less pass through the $n + 1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n + 1$ data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$. Then,

$$R_n(x) = P_n(x) - Q_n(x)$$

Since $P_n(x)$ and $Q_n(x)$ pass through all the $n + 1$ data points, we can say,

$$P_n(x_i) = Q_n(x_i) \quad \forall i \in \{0, \dots, n\}$$

Hence

$$\begin{aligned} R_n(x_i) &= P_n(x_i) - Q_n(x_i) \quad \forall i \in \{0, \dots, n\} \\ \Rightarrow R_n(x_i) &= 0 \end{aligned}$$

The n th order polynomial $R_n(x)$ has $n + 1$ zeros. A polynomial of order n can have $n + 1$ zeros only if it is identical to a zero polynomial, that is,

$$\begin{aligned} R_n(x) &\equiv 0 \\ \Rightarrow P_n(x) &\equiv Q_n(x) \end{aligned}$$

So, $P_n(x)$ and $Q_n(x)$ must be the same unique polynomial, which contradicts our initial assumption. $\square$ Q.E.D.

Rubric:

- 1 point for correctly defining $P_n(x)$, $Q_n(x)$, and $R_n(x)$.
- 1 point for showing $R_n(x)$ is a zero polynomial.
- 1 point for the final conclusion.